

2D INTERACTIVE GAME DOCUMENTATION

Game Title	Space Tunnel Shooter
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Unity Version	Unity 6.4

1. Summary of the Game Concept

Space Tunnel Shooter is a 2D side-scrolling aircraft survival shooter developed in Unity. The player controls an aircraft, avoids moving obstacles, collects health packs, shoots projectiles, and destroys enemies while trying to survive for as long as possible. Score is earned by safely passing through obstacle gaps, and the challenge increases gradually because overall game speed rises over time.

The project was designed to satisfy the assignment requirements while also giving the game a more polished feel through custom background art, scrolling background motion, audio feedback, health-color UI, and hit/heal flash effects.

2. Description of Core Features and Implementation

Player movement

The aircraft is controlled using keyboard input through a PlayerController script. Movement is handled with Rigidbody2D, and bounds are applied so the player stays inside the visible gameplay area.

Health system

A PlayerHealth script stores maximum and current health. Health is reduced when the player touches obstacles or enemies, and the UI updates instantly through a health slider and text value.

Score system

A GameManager stores the score. Each obstacle pair contains a ScoreZone trigger in the gap, and score increases only when the player passes through that trigger successfully.

Health packs

Health packs spawn at intervals and move from right to left. On contact, they call the heal function, restore health up to the maximum limit, and destroy themselves. Uncollected packs are also removed after passing the player.

Projectile attack

The player shoots using left mouse click. Projectiles are instantiated from a fire point near the aircraft nose, move forward, and self-destruct after a short time or on collision.

Enemies

Enemies spawn from the right, move left, damage the player on collision, and are destroyed by projectile hits. Destroying an enemy also increases score.

Increasing difficulty

A global speed value in the GameManager increases at timed intervals. Obstacles, enemies, and health packs all use the shared movement system, so the game becomes faster over time.

Game over and restart

When health reaches zero, movement stops, a game over panel appears, the game-over sound plays, and the player can restart the scene using the restart key.

3. Screenshots of Gameplay

The screenshots below show the working gameplay view, including the player aircraft, UI, background, obstacles, pickups, and the in-game environment.

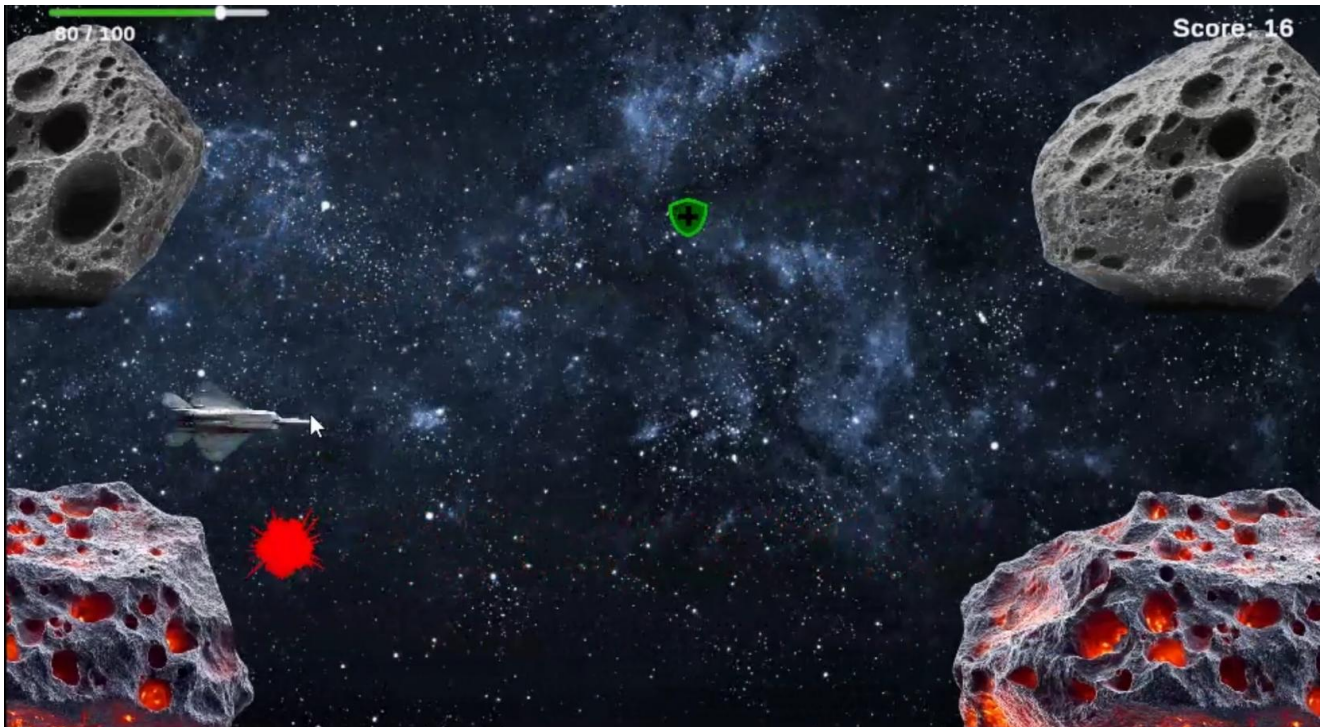


Figure 1. Gameplay screenshot from the developed Unity project.

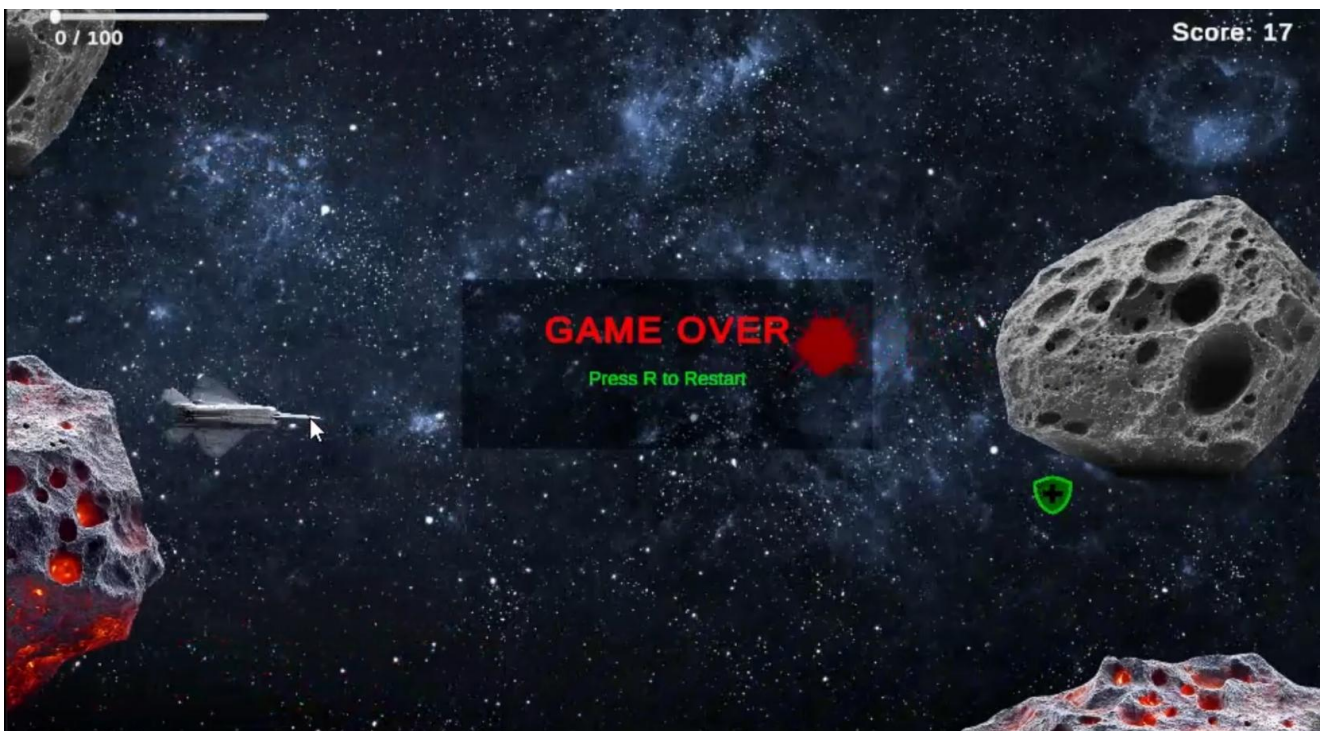


Figure 2. Gameplay screenshot from the developed Unity project.

4. Control Guide

Input	Action
W / A / S / D or Arrow Keys	Move the aircraft
Left Mouse Click	Shoot projectile
R	Restart the game after game over

5. Description of the Creative Feature

Several simple but effective creative improvements were added to make the game feel more alive and polished. The health bar changes smoothly from green to red based on remaining health, giving immediate visual feedback. The aircraft flashes red when damaged and green when healed. Improve the overall presentation of the game.

6. Credits for External Assets and Tools Used

- Unity 6.4 for 2D game development and build creation
- Visual Studio / VS Code for C# scripting
- Custom or externally sourced aircraft, obstacle, background, and health pack images used during development
- Background music and sound effects used for gameplay feedback

7. Video Link

Gameplay Video Link: [Assignment 02 GD](#)